

CHEMISORPTION OF CO ON THE Pt(111) SURFACE

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Received 8 November 1976; manuscript received in final form 11 January 1977

The system CO/Pt(111) was studied by means of LEED, thermal desorption and work function measurements. At 170 K a $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure at $\theta = 1/3$ is continuously transformed with increasing coverage into a $c4 \times 2$ structure at $\theta = 1/2$ and finally into a hexagonal close-packed layer with saturation at about $\theta = 0.68$. Due to a decrease of the adsorption energy by about 4 kcal/mole at $\theta = 0.5$ ($=7.5 \times 10^{14}$ molecules/cm²) adsorption is completed at this coverage at room temperature. The initial adsorption energy is about 33 kcal/mole. A strong tendency for disordering far below room temperature manifests itself with the LEED patterns and with the $\Delta\phi$ data. The work function at first decreases, exhibits a (temperature-dependent) minimum at $\theta = 1/3$, attains nearly the value of the clean surface at $\theta = 1/2$ and again exhibits a second (shallow) minimum around $\theta = 0.6$. A detailed discussion reveals that the observed effects may be explained by assuming the occupation of two different adsorption sites at $\theta \leq 0.5$ with different dipole moments (presumably bridge (A) and threefold coordinated (B)) whose adsorption energy differs by only 0.5 kcal/mole. At low temperatures at $\theta = 1/2$ sites A and at $\theta = 1/3$ sites B are preferentially occupied whereas their small energy difference favours disordering at increasing temperature.

1. Introduction

The adsorption of CO on platinum surfaces has been investigated very extensively since the early work of Langmuir [1]. Whereas single crystal studies with the hexagonal close-packed faces of the platinum metals Pd [2–4], Ir [5,6], Rh [7] and Ru [8] as well as of other transition metals such as Ni [3,10] and Cu [11] revealed quite similar structural properties of the observed ordered overlayers so far reports on the system CO/Pt(111) have been quite contradictory and did not fit into the general picture as established for the mentioned other elements: In an early work Morgan and Somorjai [12] reported the formation of a faint LEED spot from a $c4 \times 2$ structure after adsorption at room temperature whereas this result however could not be reproduced in a later work from the same laboratory where a 2×2

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structure was observed [13]. More recently Lambert and Comrie [14] were unable to observe the formation of an ordered adsorbate structure at all at room temperature and ascribed the different previous findings to the disturbing influence of the electron beam used for LEED.

The principal aim of the present work was to investigate whether the surface configuration of CO adsorbed on Pt(111) is in fact exceptional. During the course of this work it was found that this is not the case but that interesting ordering phenomena are occurring whose analysis could be used for characterizing the nature of different adsorption sites as well as of their difference in binding energy. After completion of the experiments we became aware of independent work on this system in other laboratories. Horn and Pritchard [15] investigated the adsorption of CO on Pt(111) by means of infrared reflection spectroscopy and work function measurements. They suggested a model for the surface configuration of the adsorbed layer which is fully supported by the conclusions drawn from the present measurements. Infrared measurements were also performed by Bradshaw and Hoffmann [43]. Froitzheim et al. [42] studied the surface vibrations of CO on Pt(111) at 320 K by means of high resolution electron energy loss spectroscopy. Flash desorption spectra for CO/Pt(111) were recorded by McCabe and Schmidt [40] and by Collins and Spicer [41]. Using recrystallized platinum ribbons (exhibiting preferred (111) orientation) photoemission and flash desorption spectra as well as work function data were taken by Collins et al. [39], whereas Shigeishi and King [16] performed infrared and thermal desorption measurements. As far as direct comparison of the data is possible the agreement is rather good, although it turned out in the meantime that recrystallized polycrystalline Pt surfaces may differ in some respects considerably from "perfect" (111) planes.

2. Experimental

The experiments were performed within a standard UHV system with a base pressure of 5×10^{-11} Torr which was equipped with 3-grid LEED optics, a cylindrical mirror analyser for Auger spectroscopy, a quadrupole mass spectrometer for thermal desorption studies and with facilities for measuring work function changes by means of the vibrating condenser technique using oxidized tantalum as material for the reference electrode.

The sample consisted of a 0.5 mm thick slice with a $6 \times 10 \text{ mm}^2$ area which was mechanically polished and spot-welded between two tungsten wires which themselves were fixed to two Mo rods mounted on the axis of the crystal manipulator. A cold-finger allowed to decrease the sample temperature to about 170 K. The temperature was measured by means of a chromel–alumel thermocouple spot-welded to the rear of the sample. Cleaning of the surface was achieved by prolonged argon ion bombardment at 600 K. Remaining spurious amounts of carbon were finally removed by brief reaction with oxygen at 800 K. Finally the sample was annealed

at about 1300 K in order to restore the clean surface periodicity. This cleaning procedure had to be repeated several times during the course of the experiments and particular care had to be taken with respect to the accumulation of carbon on the surface which was easily formed from CO in the residual gas atmosphere under the influence of the electron beam used for Auger spectroscopy.

Experiments with the electron beam as applied with the LEED observations (current density about 10^{-4} A/cm²) revealed that noticeable distortions of the adsorbed layer occurred with the electron energies above 100 eV which is in agreement with the findings of several other authors [14,17,18]. Horn [17] estimated that at 1500 eV the above mentioned current density of 10^{-4} A/cm² disrupts the adsorbed CO layer completely within about 6 min. However, in the low energy range around 50 eV it was found that these disturbing effects were negligible over those periods of time which were necessary for recording the LEED patterns. Partial dissociation of the adsorbed CO molecules was moreover found to sensitively affect the values of the work function changes $\Delta\phi$ so that even greater care had to be taken with this type of measurements. In addition thermally activated dissociation was observed to take place at elevated temperatures (≥ 600 K). As was recently demonstrated by Mason et al. [19] this process readily takes place on Pt(111) even at room temperature at structural irregularities such as steps and kinks so that the presence of spurious amounts of carbon on the surface cannot be completely ruled out. As will be discussed later this effect is believed to be responsible for the slight difference between our and Horn's [17] $\Delta\phi$ data. Since however the surface concentration of this element was always below the detection limits of AES there is no indication that it affects all the other conclusions of the present work.

3. Results

3.1. Low energy electron diffraction

If a clean Pt(111) surface is exposed at 300 K to CO after about 1 L rather diffuse diffraction spots from a $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure appear in the LEED pattern. These features are nearly not visible at room temperature but become more intense and sharper if the temperature of the sample is lowered. At the lowest attainable temperature of 170 K the "extra" spots were, however, still relatively broad indicating still a high degree of disorder in the adsorbed layer. Further increase of the exposure leads to a continuous splitting of these spots in a manner identical to that observed for example with CO adsorbed on Pd(111) [2] or Ni(111) [9]. After about 2 L a pattern characterizing a $c4 \times 2$ structure is observed (fig. 1a) whose diffraction spots remain rather bright and sharp up to temperatures of about 270 K, but again decrease in intensity with further increase of the temperature and finally disappear around 350 K. This effect is not due to desorption since cooling the sample in vacuo causes the reversible reappearance of the $c4 \times 2$ pattern again, indicating the operation of an order-disorder transition.

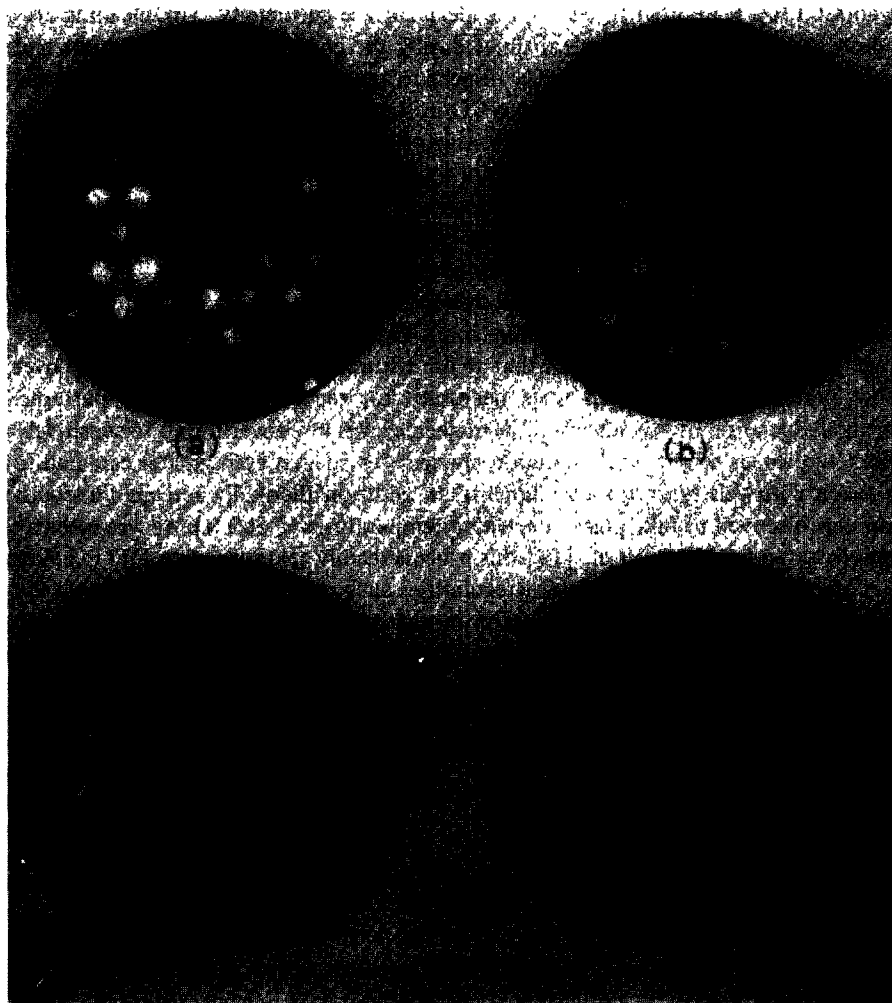


Fig. 1. LEED patterns from ordered CO overlayers at 170 K ($U = 38$ V): (a) $c4 \times 2$ structure at $\theta = 0.5$; (b) Metastable structure at $\theta \approx 0.6$ with elongated diffraction spots; (c) Equilibrium structure at $\theta \approx 0.6$; (d) Saturation with $\theta \approx 0.7$.

A further increase of the coverage beyond that characteristic for the $c4 \times 2$ structure is only possible if the sample temperature is lowered considerably below room temperature. These findings indicate a steplike decrease of the adsorption energy at this surface concentration. Additional CO uptake (at a much smaller rate) was observed to take place at 170 K leading to LEED patterns as reproduced in figs. 1b–d.

Fig. 1b represents a metastable surface configuration characterized by slightly

elongated spots which was observed immediately after increasing the exposure beyond that necessary for the formation of the $c4 \times 2$ pattern. Keeping the sample *in vacuo* after a few seconds this pattern transforms into one with smaller and round spots of the type as reproduced in fig. 1c. Obviously surface diffusion which is a necessary prerequisite for the formation of two-dimensional order is partly restricted at this temperature so that this kinetic phenomenon may be directly followed in the LEED pattern. The final stage of adsorption at 170 K is reached after considerably large exposures (≥ 200 L) and is characterized by the LEED pattern shown in fig. 1d.

A detailed inspection of the LEED patterns revealed that these were representatives for a continuous uniaxial compression of the unit cell of the adsorbate structure along the $[110]$ direction with increasing coverage as drawn schematically in fig. 2 in a similar manner as observed with the (111) surfaces of other fcc metals [2,4,9–11]. The $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure accordingly corresponds to a coverage $\theta = 1/3$. The primitive cell of the $c4 \times 2$ structure is denoted in matrix notation [20] by $\begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}$ and corresponds to $\theta = 0.5$. The unit cell of the structure whose LEED pattern is reproduced in fig. 1c is denoted by $\begin{pmatrix} 4/3 & 1/3 \\ 1/3 & 4/3 \end{pmatrix}$ and corresponds to $\theta = 0.6$. The maximum coverage is about $\theta \approx 0.68$ or about 1.05×10^{15} molecules/cm².

Above $\theta = 0.5$ order–disorder phenomena were not observed to interfere with the establishment of reversible adsorption–desorption equilibrium and therefore the LEED patterns depend uniquely on the coverage. This is ascribed to the operation of short-range repulsive interactions between neighbouring adsorbed particles which prevent them from large displacements at higher surface concentrations. The

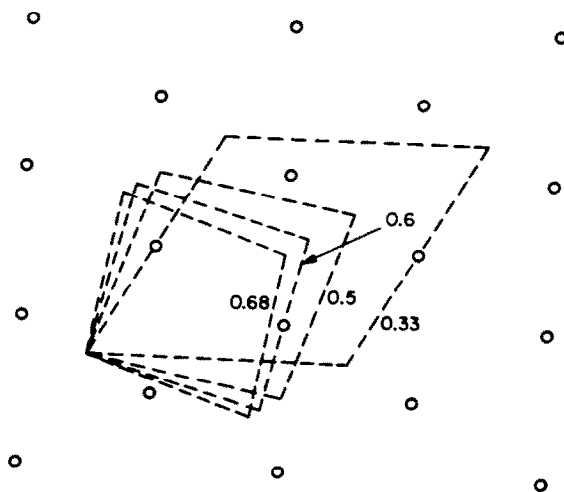


Fig. 2. Primitive unit cells for ordered adsorbate structures at different coverages as derived from the LEED patterns.

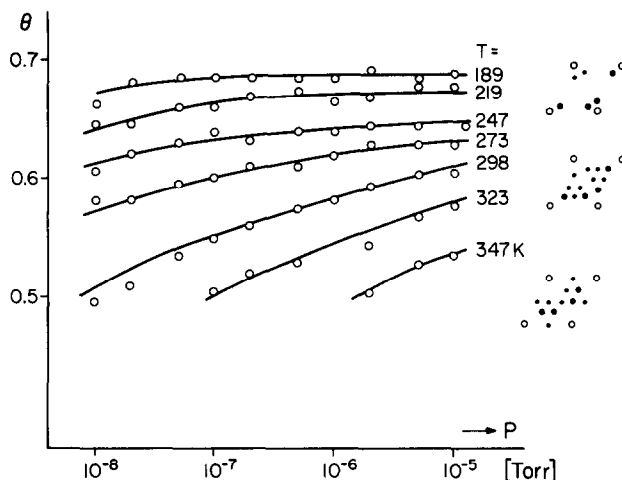


Fig. 3. Adsorption isotherms for $\theta > 0.5$ at different temperatures as evaluated from LEED observations.

LEED patterns therefore could be used to determine the equilibrium coverages ($\theta > 0.5$) as a function of temperature at fixed pressures. Such a procedure was applied for the first time by Tracy and Palmberg [21] with the system CO/Pt(100). The resulting adsorption isobars are plotted in the form of isotherms in fig. 3. From these data isosteric heats of adsorption were derived by applying the Clausius–Clapeyron equation which are included in fig. 4. The accuracy of these determinations is ± 1 kcal/mole, but becomes worse near saturation.

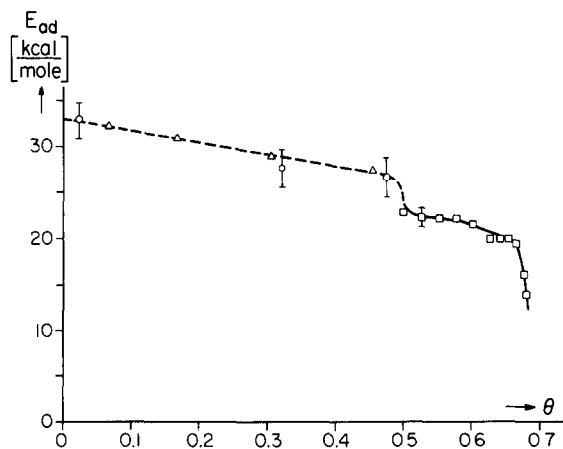


Fig. 4. Adsorption energy E_{ad} as a function of coverage: ($\circ$) isosteric heats derived from work function data; (Δ) some of the values obtained from thermal desorption spectra assuming $\nu = 10^{15} \text{ sec}^{-1}$; ($\square$) isosteric heats derived from LEED data.

3.2. Thermal desorption

A series of thermal desorption spectra recorded with a heating rate of 15 K/sec is reproduced in fig. 5. Unfortunately our cooling device did not allow rapid heating from below room temperature and therefore prevented measurements for $\theta \geq 0.5$. On the high-temperature side of the main peak a second small maximum appeared which was saturated at relatively small exposures. This feature could be isolated from the rest by proper heating of the sample and revealed neither to influence the LEED pattern nor the work function of the clean front plane of the sample. It was therefore concluded that it arises either from the crystal holder or from other parts of the sample. Similar observations were made by McCabe and Schmidt [40] and by Bonzel [22]. In a systemic study with smooth and stepped Pt(111) surfaces Collins and Spicer [41] demonstrated clearly that this high-temperature state arises from adsorption at step sites. For quantitative analysis of the spectra in the present work this contribution was therefore neglected. It is interesting to note that with both TDS studies with recrystallized Pt ribbons [16,39] the main desorption maximum is located above 500 K (i.e. corresponding to the present high-temperature state) indicating pronounced energetic difference to the (111) single crystal surfaces.

The TDS maximum shifts to somewhat smaller temperatures with increasing coverage. Since adsorption occurs associatively desorption kinetics has to be ascribed to a continuous slight decrease of the adsorption energy with increasing coverage, which would be in agreement with the analysis of McCabe and Schmidt [40]. Our spectra reveal no discontinuity at $\theta = 1/3$ where compression of the $\sqrt{3} \times \sqrt{3}/R30^\circ$

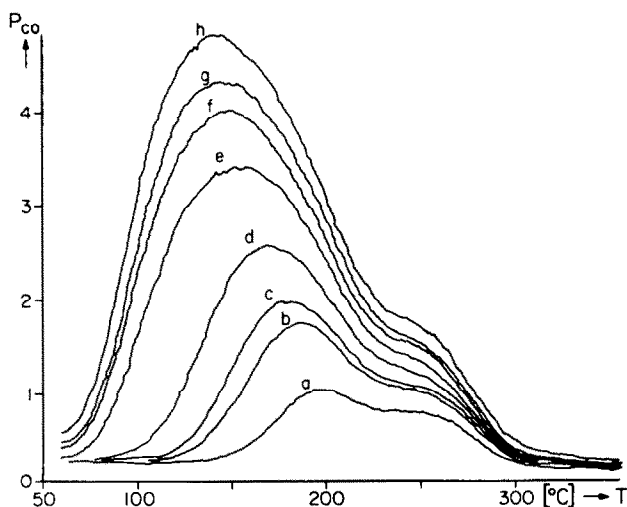


Fig. 5. Thermal desorption spectra after different CO exposures at 300 K. Coverages: (a) 0.07, (b) 0.14, (c) 0.18, (d) 0.24, (e) 0.33, (f) 0.38, (g) 0.40, (h) 0.45.

structure starts. With Pd(111) at this point a shoulder on the low-temperature side of the main desorption peak started to grow indicating a steplike decrease of the adsorption energy which becomes even more pronounced with the isosteric heat of adsorption as derived from isotherm data [2]. The reason for a different behaviour in the present case is caused by the fact that thermal desorption occurs at temperatures far above the ordering temperature for the $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure, i.e. it takes place from a random surface configuration of the adsorbed molecules on sites with (slightly) varying binding energy. The continuous decrease of the adsorption energy with coverage is best explained in terms of long-range repulsive interactions between the adsorbed particles which continuously increase as the mean nearest-neighbour distance decreases. This point becomes even more clear from an inspection of the infrared data by Pritchard and Horn [15] recorded with varying coverages at 80 and 300 K which will be briefly discussed later. However, analysis of the adsorption isotherm—work function data in terms of the isosteric heat of adsorption indicates that this quantity remains constant at low coverages. This apparent discrepancy between the results of equilibrium and kinetic measurements is probably due to ignoring the role of the “precursor” state in the analysis of thermal desorption spectra as suggested by King [45]. Unfortunately the present adsorption energies derived from $\Delta\phi$ data are not accurate enough to decide this interesting question.

No attempt was made to derive the activation energies E_d for desorption from a full line-shape analysis of the thermal desorption spectra [23] but instead only estimates on these quantities were made by applying Redhead's formula [24] correlating E_d with the TDS peak temperature T_p for the first-order kinetics through

$$E_d = RT_p [\ln(\nu T_p/\alpha) - \ln(\ln \nu/\alpha)] \quad (1)$$

α is the heating rate and ν the pre-exponential factor for desorption. Since adsorption is not activated E_d should be equal to the adsorption energy E_{ad} . If (as usually) ν is assumed to be 10^{13} sec^{-1} E_d extrapolated to zero coverage results to be around 28 kcal/mole. By applying a similar analysis (with $\nu = 10^{13} \text{ sec}^{-1}$) Collins and Spicer [41] arrived at a number of 27 kcal/mole and McCabe and Schmidt [40] at 29.6 kcal/mole. In the work with recrystallized Pt ribbon desorption energies at low coverages of 37 [16] and 32 kcal/mole [39] were derived (again assuming $\nu = 10^{13} \text{ sec}^{-1}$), confirming the above made conclusion on the energetic difference between these surfaces and Pt(111). In view of the derived isosteric heats (from $\Delta\phi$) as described in the next section there is strong evidence that the present value of 28 kcal/mole is somewhat too small. This conclusion is not singular for the present case but seems to be quite common for CO adsorption. As discussed in detail by Comrie and Weinberg [5] with this adsorbate ν has to be assumed to be generally considerably higher in order to obtain agreement with the isosteric heat of adsorption, e.g. $2 \times 10^{14} \text{ sec}^{-1}$ for Ir(111) and $9 \times 10^{15} \text{ sec}^{-1}$ for Pd(111). With the present system reasonable values for E_d (i.e. agreement with isosteric heats) are achieved with $\nu = 10^{15} \text{ sec}^{-1}$ and some of the data thus resulting are included in fig. 4.

Calibration of the coverage was achieved from the area of the TDS trace characterizing saturation at room temperature which was on the other hand identified with $\theta = 0.5$ ($n_s = 7.6 \times 10^{14}$ molecules/cm²) from the corresponding LEED pattern. From a quantitative analysis of thermal desorption data Shigeishi and King [16] arrived at room temperature at a saturation concentration of $\sim 7 \times 10^{14}$ molecules/cm². This rather good agreement is a further support for the correct interpretation of the observed LEED patterns.

3.3. Variation of the work function

The work function of the sample was observed to vary with continuous CO uptake in a peculiar way. Fig. 6 shows a typical recorder trace of $\Delta\phi$ as a function of exposure at 190 K. The work function at first decreases, then passes through a pronounced minimum, reaches a maximum and then decreases again and passes through a second shallow minimum. The purpose of this figure is just to illustrate qualitatively the shape of the work function variation as a function of exposure as recorded in a single experiment. However it does not represent true equilibrium values for $\Delta\phi$ as a function of coverage. It was observed that attaining the surface equilibrium configuration after a certain gas exposure was accompanied by some relaxation caused by surface diffusion. As a consequence the $\Delta\phi$ minimum is not so deep as it would be under equilibrium conditions. Also the height of the maximum was observed to be influenced slightly by the rate of adsorption, under equilibrium conditions it was observed to have almost exactly the value $\Delta\phi = 0$ eV. In addition with exposures exceeding those for establishing the $\Delta\phi$ maximum equilibration of the adsorbed layer (and therefore of $\Delta\phi$) was observed sometimes to be rather sluggish (as indicated already by the high coverage LEED observations) and it revealed to be rather difficult to attain reproducible $\Delta\phi$ versus exposure data in this range.

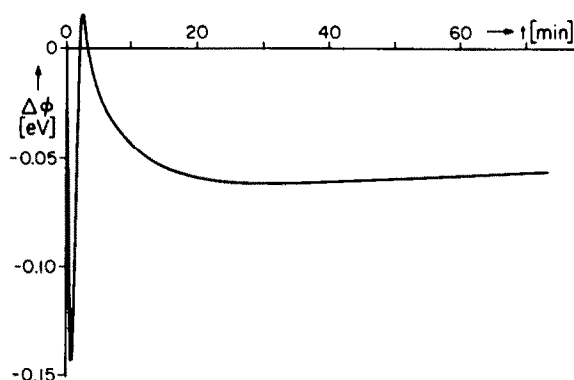


Fig. 6. Variation of the work function $\Delta\phi$ with time at $p_{\text{CO}} = 10^{-7}$ Torr and $T = 190$ K.

By using the calibration of θ versus exposure as derived from LEED and TDS data it was found that the first minimum occurs almost exactly at $\theta = 1/3$ and the maximum at $\theta = 1/2$. The second (shallow) minimum is associated with $\theta \approx 0.65$ although in this range the calibration is relatively uncertain. A plot of $\Delta\phi$ as a function of coverage at 323 K is reproduced in fig. 7.

Interestingly, the depth of the (first) minimum, $\Delta\phi_{\min}$, was found to vary considerably with temperature but on the other hand to occur always at the same coverage $\theta = 1/3$. Much less variation with temperature was found to take place with $\Delta\phi$ of the maximum which always occurs at $\theta = 1/2$ and which to a good approximation may be regarded to be constant, $\Delta\phi_{\max} \approx 0$ eV, in the following discussion. These findings are in qualitative accordance with the observations by Horn and Pritchard [15] whose data for $\Delta\phi_{\min}(T)$ are plotted in fig. 10. The absolute values of these authors are slightly larger than ours which is believed to be caused by tiny amounts of carbon deposits on the surfaces used with the present investigations. The conclusions are not affected by this slight difference but since Horn's data cover a much wider range of temperatures these will be used for quantitative discussion of the structural model proposed in section 4. The work function of the clean Pt(111) surface was found to vary only very slightly with temperature ($d\phi/dT = -3 \times 10^{-5}$ eV/K) so that this effect is completely negligible in the later discussion. With a recrystallized Pt ribbon Collins et al. [39] observed only a continuous increase of the work function whereas recent measurements with Pt(111) also exhibit a pronounced minimum [44].

Since $\Delta\phi$ revealed not to be a function of coverage *alone* but also of temperature, it was not possible to derive very accurate data for the isosteric heats of adsorption from the measurement of adsorption isotherms $\Delta\phi = f(p)$ at different temperatures. However it was found that above 300 K the influence of temperature on $\Delta\phi$ (at constant coverage) could fairly well be eliminated so that reliable values (± 2 kcal/mole) at $\theta \approx 0$ (33 kcal/mole) and at $\theta = 1/3$ and slightly below $\theta = 1/2$ (27 kcal/

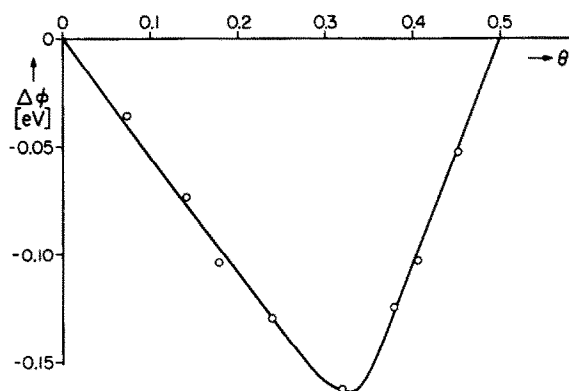


Fig. 7. Variation of the work function with coverage at $T = 323$ K.

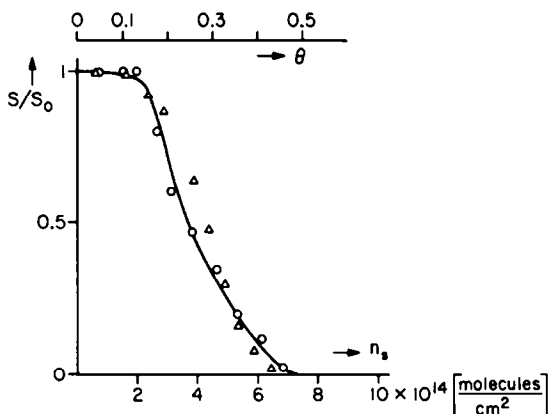


Fig. 8. Variation of the relative sticking coefficient s/s_0 at 300 K with coverage θ resp. surface concentration n_s : (○) this work; (△) data of Shigeishi and King [16].

mole) were obtained. Unfortunately no overlap of data for the isosteric heats derived by two different methods ($\Delta\phi$ and LEED) was possible since LEED measurements could only be performed above $\theta = 0.5$ where on the other hand the $\Delta\phi$ data were not reproducible enough. But all the qualitative observations indicate that there is a steplike decrease of E_{ad} at $\theta = 1/2$ upon the onset of the compression of the $c4 \times 2$ unit cell. We are further convinced that the E_{ad} values derived from $\Delta\phi$ isosteres are reliable enough to justify the assumption of a preexponential larger than 10^{13} sec^{-1} for the analysis of the thermal desorption spectra in order to achieve agreement with both sets of data.

Information on the kinetics of adsorption was obtained by measuring $\Delta\phi$ as a function of exposure and by using the calibration of exposure versus coverage from the LEED and TDS data. The initial sticking probability s_0 was estimated to be not far away from unity but could not be determined with great accuracy due to imperfect calibration of the *absolute* pressure measurements. Shigeishi and King [16] arrived at $s_0 = 0.67$ which supports our conclusion. Their data on the variation of the relative sticking coefficient s/s_0 with surface concentration are reproduced in fig. 8 together with the data determined in the present work. The agreement is very good concerning the shape of this curve as well as the calibration of the coverage scale which was obtained in different ways. McCabe and Schmidt [40] derived $s_0 = 0.45$ and found again a quite similar shape for the variation of the sticking coefficient with coverage. The shape of the s/s_0 versus θ curve indicates the operation of a "precursor" state which is frequently observed with CO adsorption on transition metals [25]. However a detailed inspection reveals that in the present case no adequate description is achieved with the simple Kisliuk model in contrast to the conclusions of Comrie [26].

4. Discussion

The initial adsorption energy of CO on the hexagonal planes of group VIII metals was found to be 26.5 kcal/mole for Ni [3], 29 kcal/mole for Ru [8], 34 kcal/mole for Ir [5,6] and Pd [2]. The present data for Pt(111) thus fit well into this scheme. The same holds for the maximum density of adsorbed particles which was found to be 1.05×10^{15} molecules/cm²: About the same number was derived for Ni(111) [10], Ru(001) [8], Ir(111) [5,6] and in a recent low temperature study also for Pd(111) [4]. This value appears also to be rather independent of the crystallographic orientation of the surface since it also holds, e.g., for Ni(100) [4] and Ni(110) [28] as well as for Pt(110) [26] and is obviously determined by the tendency for formation of a close-packed layer of the adsorbed CO molecules. Even with polycrystalline evaporated Pt films Stephan and Poncet [29] recently arrived at maximum surface densities between 0.8 and 1.2×10^{15} molecules/cm² which are of the same order of magnitude. The same sequence of overlayer structures as for CO/Pt(111) with continuous compression of the unit cell was for example also observed with Ni(111) [9] and Pd(111) [2] surfaces. In the present case, however, there is a strong tendency for disordering at coverages below 1/2 which is believed to be the main reason for the fact that previous LEED investigations at room temperature failed to detect clear evidence for the formation of ordered structures. It is interesting to note that also with CO/Ir(111) [5] as well as with CO/Ru(001) [8] some indication was found for an order-disorder transition although in these cases ordered overlayer structures are still clearly discernible at room temperature. A second and even more pronounced difference concerns the sign of the work function variation and the peculiar shape of the $\Delta\phi(\theta)$ curve. With all the other group VIII metals investigated so far CO adsorption causes a monotonic *increase* of the work function. A work function increase by 0.3 eV was recently also reported for polycrystalline Pt [39]. A *decrease* and the formation of a minimum as with Pt(111) was on the other hand also observed with Cu(111) [30]. Similarities between these two latter systems were also observed with respect to the frequency of the C—O stretching vibration as detected by means of infrared reflection spectroscopy [15,16]: Only a single absorption band around 2100 cm⁻¹ is observed which is commonly referred to "linear" bond species whereas no signal in the 1800–2000 cm⁻¹ (as present with other group VIII metals and identified with "bridge" bonds) could be detected.

According to the generally accepted model for CO chemisorption [31,32] bond formation takes place mainly through coupling of the 5 σ orbital of CO to the sp bands of the metal and back-donation of metallic d electrons into the 2 π^* orbitals. With respect to the C—O bond the 5 σ orbital is essentially non-bonding whereas the 2 π^* orbitals are anti-bonding. In this picture the back-donation effect therefore will be responsible for a lowering of the C—O stretching frequency and eventually for a net electronic charge on the adsorbate leading to an increase of the work function. Thus with CO adsorbed on Ni and Pd ν bands around 1800 cm⁻¹ as well as an increase of the work function by about 1 eV is observed. In contrast, with Cu and Pt

the work function even decreases and the CO vibration frequency is only slightly displaced from its value in the gas phase. It has thus to be concluded that with these metals back-donation of d electrons into empty adsorbate orbitals is much less pronounced than with the other group VIII elements. This effect has been treated theoretically for Cu leading to a correct prediction of the lower adsorption energy as compared with Ni [32]. No calculations have been performed so far for Pt(111). In this case an essential difficulty arises from the fact that the adsorption energy is practically identical to that found with Ir(111) or Pd(111) which only may be explained by assuming that the loss of back-donation is compensated by a stronger coupling of the 5σ level to the metal. As a consequence one would expect that the energy separation between the 5σ and 1π level of CO adsorbed on Pt(111) should be somewhat smaller than on Ir(111) or Pd(111). Although UPS data are available for these three systems [4,19,33], this technique is not sensitive enough to yield clear evidence for the postulated effect. It is interesting to note that also in the case of hydrogen adsorption the Pt(111) surface exhibits an exceptional behavior: With this system again the work function was observed to decrease [34] whereas it increases on Ni(111) [35] as well as on Pd(111) [36]. However, the situation becomes more complicated if the recent electron energy loss experiments of Froitzheim et al. [42] are taken into account: In agreement with the i.r. measurements they observed only a single absorption band around 2100 cm^{-1} at low coverages, but in contrast also with increasing exposure the emergence of a second band at 1870 cm^{-1} . For further discussion the adsorption site associated with the 1870 cm^{-1} band will be designated by "A", that at 2100 cm^{-1} by "B". Froitzheim et al. [42] identified the A sites with bridge positions and the B sites with linearly adsorbed species, in the usual manner. Without knowing these results we had postulated a 2 site model as outlined below, where the A sites are again bridge positions whereas the B sites are three-fold coordinated. The problem with Ibach's assignment consists in the fact that the B sites are obviously energetically more favourable (which one would expect for a multi-centered site from general experience [32]) whereas with all other known systems where linear and bridge species coexist the latter are more tightly bound. On the other hand our model cannot explain why the threefold coordinated adsorbate should exhibit a higher wavenumber for the C—O stretching vibration than the bridge species. A definite solution of this interesting problem can only be expected from a quantitative structure analysis but is on the other hand not of decisive importance for the general conclusions from the following discussion.

The occurrence of a minimum with the $\Delta\phi(\theta)$ curve might probably be due to mutual depolarization effects between the adsorbed molecules. But in this case the maximum at $\theta = 1/2$ followed by a new decrease as well as the temperature dependence of $\Delta\phi_{\min}$ could hardly be explained. In fact Horn's [17] infrared data at 80 K exhibit up to a coverage corresponding to the work function minimum ($\theta = 1/3$) no shift but only a continuous increase of the intensity of the absorption band indicating negligible depolarization and identical adsorption sites under these conditions. A continuous shift of the ir band was only observed above $\theta = 1/3$ which coincides

with the onset of compression of the adsorbate unit cell, i.e., displacement from the low-coverage adsorption sites, as revealed by the LEED observations. At 300 K even below $\theta = 1/3$ the ir band is shifting with coverage so that it is concluded that the adsorbed molecules are no longer distributed over equivalent adsorption sites which agrees with the observed disordering of the adsorbate layer as becomes evident from the degradation of the LEED pattern. The ELS data by Froitzheim et al. [42] taken at 320 K indicate that at this temperature about equal numbers of A and B sites are occupied.

Although dipole–dipole coupling may play an important role for the explanation of the infrared observations at high coverages [15,16] the arguments mentioned beforehand justify an interpretation of the $\Delta\phi$ data primarily in terms of structural properties of the adsorbate layer. Using the structural information from LEED (including the observed tendency for disordering with rising temperature) the shape of the $\Delta\phi(\theta)$ curve as well as the observed variation of $\Delta\phi_{\min}$ with temperature are explained as follows: The (111) plane offers three types of highly symmetric adsorption sites, namely “on top” (single site), in twofold (bridge) position and with three-fold coordination. Quantum chemical calculations for Ni and Pd [32] revealed that the latter two are energetically slightly favoured over other positions on the surface with nearly no difference in adsorption energy between the twofold and three-fold sites. Although no results are available for platinum the general features are believed to be similar with this metal. By applying the empirical BEBO-model to CO/Pt(111) Weinberg [37] suggested that with this surface the adsorption energy be rather constant (about 28 kcal/mole) over the whole plane with a slight preference for the threefold coordinated sites.

At $\theta = 0.5$ the only possibility to bring the CO molecules into equivalent (low energy) positions which can be reconciled with the primitive mesh of the $c4 \times 2$ structure is given by adsorption on “bridge” sites (A). The corresponding structure model is reproduced in fig. 9b). A CO molecule adsorbed in such a “bridge” position obviously causes only a negligible change of the work function, that means this type of adsorbate complex exhibits a vanishing dipole moment, $\mu_A \approx 0.0$ Debye. At $\theta < 1/3$ (and low enough temperatures) obviously not the bridge sites are occupied since at first $\Delta\phi$ decreases continuously indicating a positive dipole moment. It is assumed that sites with threefold coordination (B) are energetically slightly favoured and thus at first occupied until at $\theta = 1/3$ the observed $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure is completed whose structure model is reproduced in fig. 9a. Completion of this structure just coincides with the occurrence of the work function minimum at $\theta = 1/3$. At 80 K Horn [17] determined $\Delta\phi_{\min} = -0.37$ eV. As will be demonstrated below at this temperature and coverage nearly all adsorbate molecules will be on type B sites for which kind of complex therefore a (positive) dipole moment $\mu_B = 0.2$ Debye is evaluated. Above $\theta = 1/3$ the continuous compression of the adsorbate unit cell causes an increasing number of triply bound CO molecules to be displaced into positions exhibiting a smaller dipole moment until finally at $\theta = 1/2$ all particles are in bridge positions and $\Delta\phi$ becomes nearly zero. Similar but less pronounced

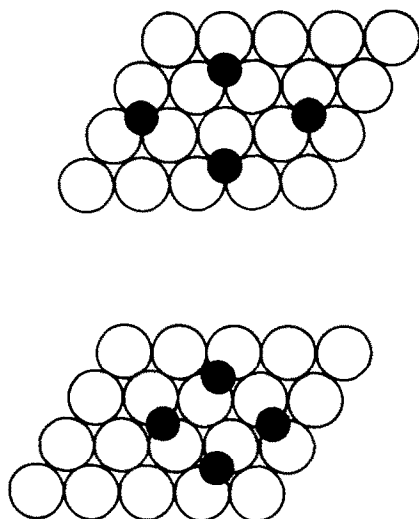


Fig. 9. Structure models for CO adsorbed on Pt(111): (a) $\theta = 1/3$, $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure with preferred occupation of threefold coordinated sites (B sites); (b) $\theta = 1/2$, $c4 \times 2$ structure with adsorption on "bridge" positions (A sites).

effects obviously come into play above $\theta = 0.5$ where the compression of the unit cell continues so that the adsorbed molecules now are partly displaced from the "bridge" positions (with minimum dipole moments) leading to the second (shallow) minimum of the $\Delta\phi(\theta)$ curve.

The arguments presented so far strictly hold only for $T = 0$ K where at $\theta = 1/3$ all adsorbed particles are predicted to occupy type B sites exhibiting the lowest total energy. Since however the difference in adsorption energy between A and B positions is predicted to be rather small with increasing temperature an increasing amount of adsorbed molecules should statistically also occupy the slightly less favourable bridge positions. This effect would explain the observed decrease of $|\Delta\phi_{\min}|$ with increasing temperature as well as the structural order-disorder transformation as revealed by the LEED patterns. Taking into account the ratio of numbers of triple and bridge sites on a (111) plane of 2 : 3 the following expression for the fractional coverage θ_B occupied by triply coordinated CO in thermal equilibrium at a total coverage $\theta \leq 1/3$ is derived:

$$\theta_B = \theta / [1 + \frac{3}{2} \exp(-\Delta\epsilon/kT)] . \quad (2)$$

$\Delta\epsilon$ is the difference in adsorption energy between B and A-coordinated sites. With $\mu_A = 0$, i.e., CO molecules in bridge positions do not contribute to the work function change, from this equation follows that the variation of the work function minimum $\Delta\phi_{\min}$ at $\theta = 1/3$ with temperature should obey the equation

$$\Delta\phi_{\min}(T) = \Delta\phi_{\min}^0 / [1 + \frac{3}{2} \exp(-\Delta\epsilon/kT)] . \quad (3)$$

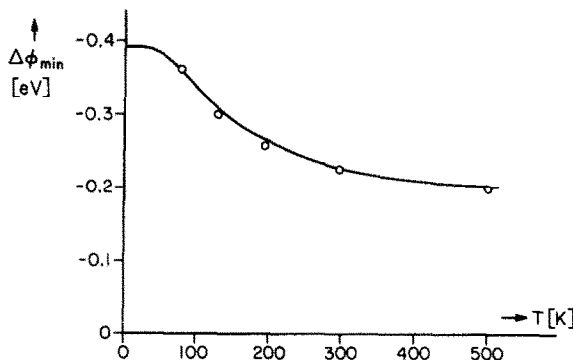


Fig. 10. Variation of the minimum of the work function change $\Delta\phi_{\min}$ at $\theta = 1/3$ with temperature. Circles: experimental data of Horn [17]. Full line: theoretical curve according to eq. (3) with $\Delta\phi^0 = -0.39$ eV, $\Delta\epsilon = 0.45$ kcal/mole.

Fig. 10 shows a plot of experimental data for $\Delta\phi_{\min}$ as a function of temperature taken from the work of Horn [17]. (As mentioned above our own data are slightly smaller and only available over a more limited range of temperatures but are in qualitatively very good agreement as far as comparison can be made.) These data are nicely approximated by the theoretical curve according to eq. (3) for which the best fit (full line of fig. 10) was obtained with $\Delta\phi_{\min}^0 = -0.39$ eV and $\Delta\epsilon = 0.45$ kcal/mole. It is thus concluded that this latter small number represents the difference in adsorption energy between threefold- and twofold-coordinated sites. It has to be mentioned that quite similar conclusions were reached intuitively by Horn and Pritchard [15] on the basis of their $\Delta\phi$ and ir data but having not the structural information from LEED.

As a consequence of this result it is predicted that even at 170 K (the lowest temperature attained during the LEED observations) at $\theta \approx 1/3$ about 30% of the adsorbed particles should be displaced from the low-energy B sites into bridge positions. As mentioned in section 3.1 in fact the LEED pattern from the $\sqrt{3} \times \sqrt{3}/R30^\circ$ structure was always rather diffuse indicating a high degree of disorder even at these low temperatures. At 320 K according to our model at $\theta = 1/3$ about 60% of the adsorbates are predicted to be in B positions and 40% on A sites. This agrees fairly well with the relative surface concentrations of these species as derived from the intensities of the two loss peaks in the work of Froitzheim et al. [42] which was performed at this temperature. The presented model assumed the unrestricted occupation of type A and B adsorption sites in thermal equilibrium. This will certainly only be possible at small enough coverages. Above $\theta = 1/3$ the higher adsorbate concentration will presumably prevent this for steric reasons. This becomes particularly evident for the close-packed structures above $\theta = 1/2$ where disordering is prevented and is preceded by desorption. Therefore we restrict the applicability of eq. (2) to the coverage range below $\theta \leq 1/3$.

The establishment of the equilibrium configuration proceeds obviously fast enough even at rather low temperatures as long as the coverage is below $\theta = 1/2$. At higher coverages kinetic phenomena may be followed by the variation of the LEED patterns as well as of $\Delta\phi$ with time. This is probably due to the fact that near saturation the mobility of a single adsorbed particle becomes restricted due to its dense surrounding by nearest neighbours so that collective displacements will become necessary which would be consistent with field emission observations by Lewis and Gomer [38] whereafter such a layer becomes mobile only around room temperature. These conclusions are not in full agreement with the statement by Shigeishi and King [16] whereafter the overlayer is not completely equilibrated at low coverages at temperatures ≤ 300 K. This is again believed to be one of the differences between recrystallized and single-crystal surfaces.

During the whole discussion the operation of interactions between the adsorbed particles were neglected. These are presumably responsible for the continuous decrease of the adsorption energy with increasing coverage and certainly play an important role above $\theta = 1/2$. The steplike decrease of E_{ad} at this coverage is ascribed to the onset of a "squeezing" of the adsorbate unit cell and is responsible for the fact that at room temperature and low CO pressures saturation is reached at $\theta = 1/2$, as well as that at low temperatures the rate of further CO uptake slows down considerably. A similar effect has for example been reported for the CO/Pd(100) system by Tracy and Palmberg [21].

Acknowledgements

We are very grateful to Drs. K. Horn and J. Pritchard for helpful discussions during this work. H. Ibach, D.A. King, L.D. Schmidt and W.E. Spicer made available to us their results prior to publication and helped us with fruitful comments. The manuscript was completed while one of the authors (G.E.) was staying as a Sherman Fairchild Distinguished Scholar at Caltech whose kind hospitality is gratefully acknowledged. Financial support of this work was obtained by the Deutsche Forschungsgemeinschaft through SFB 128.

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